DOCUMENT NAME:	FORA Communication Protocol for WS series
D/N:	Rev:
FILE NAME: TICD-WS series	Update: 2015/01/19

Revision History.

#	Date.	Done by.	The update.	New rev.
1	2010/12/08	Peano	Primary Edition.	Ver1.0
2	2012/01/09	Carol	Modification.	Ver1.01
3	2012/05/07	Carol	Modification.	Ver1.02
4	2012/07/23	Carol	Command 71, 72 code / user ID value	Ver1.03
5	2012/07/24	Carol	Modification.	Ver1.04
6	2012/09/11	Carol	Command 72.	Ver1.05
7	2013/03/27	Carol	Byte 4 in 0x71 command	Ver1.06
8	2013/06/03	Carol	Command 0x54 modification.	Ver1.07
9	2015/01/19	Carol	Command 0x71 modification.	Ver1.08
10	2015/07/02	Carol	Note for different Bluetooth versions	Ver1.09

Table Of Contents.

1. General.

1.1. Overview.

The document describes the commands which will be used to communicate with 25xx series.

Before retrieving data in the device, please make sure you have already established the proper connection channel between the device and the host side. For example, if you use USB cable, be sure that the USB driver has been properly installed; or if you use Bluetooth communication, be sure that the connection has been successfully established.

Generally speaking, the device will be the slave side waiting for the request from the host which is usually the computer or the hub/gateway.

Note for different Bluetooth versions:
For Bluetooth v2 meter: Please connect to meter by SPP (Serial Port Profile). Meter always stands by for both pairing and connection, the PIN for paring is 111111.

For Bluetooth v3 meter: Please connect to meter by SPP (Serial Port Profile). Meter must be paired before any connection, please find the instruction in meter manual for the pairing process. The meter supports SSP (Simple Secure Pairing), as a result PIN is not required. If host does not support SSP, please use 0000 as PIN if required.

For Bluetooth v4 meter: Please connect to meter by following GATT service:
UUID Base: 1212-efde-1523-785feabcd123
Service UUID: 0x1523
Characteristic: 0x1524 (write/notify)

1.2. Document Overview.

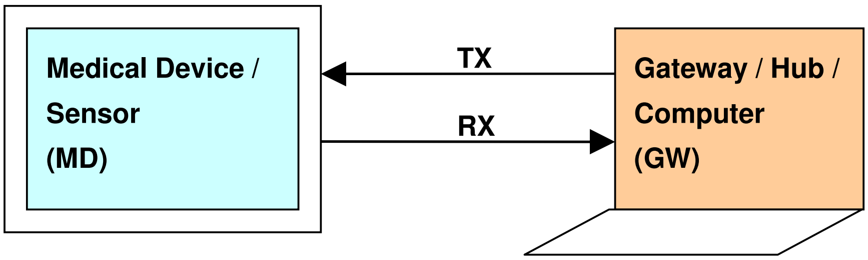
For the first time user on the device integration, please read the 2.1 frame structure. If you are familiar with serial port communication, you can review the command list table in section 2.2 directly. The following sub sections are all about the details for each command.

1.3. Definitions, Acronyms, and Abbreviations.

GW	Gateway / Hub / Computer side
MD	Medica Device, ie. blood glucose meter, blood pressure meter or scales, etc.
CMD	Command
NA	Not available

2. Interface specification

Block Diagram



The communication interface is base on RS232 serial port communication :

Baud rate	19200
Data bits	8
Parity	No
Start bit	1
Stop bit	1

2.1. Frame Structure

8-byte frame

Byte	1	2	3	4	5	6	7	8
Name	Start	CMD	Data_0	Data_1	Data_2	Data_3	Stop	ChkSum
Format	HEX							
GW→MD	51	CMD	Data_0	Data_1	Data_2	Data_3	A3	[1..7]
GW←MD	51	ACK	Data_0	Data_1	Data_2	Data_3	A5	[1..7]

Variable frame

Byte	1	2	3	4 ~ n-2	n-1	n
Name	Start	CMD	Data Length	Index, Address, or Data	Stop	Check Sum
Format	HEX					
GW→MD	51	CMD	Length	… …	A3	[1…n-1]
GW←MD	51	ACK	Length	… …	A5	[1…n-1]

ie. Check sum means the summation from byte 1 to byte n-1 with 8-bit length

2.2. Command List

Message name	CMD/ACK	Direction	Length (Bytes)	Note
Read device clock time	0x23	GW→MD	8	
	0x23	GW←MD	8	
Read device model	0x24	GW→MD	8	
	0x24	GW←MD	8	
Read device serial number, part 1	0x27	GW→MD	8	
	0x27	GW←MD	8	
Read device serial number, part 2	0x28	GW→MD	8	
	0x28	GW←MD	8	
Read storage number of data	0x2B	GW→MD	8	
	0x2B	GW←MD	8	
Write device clock time	0x33	GW→MD	8	
	0x33	GW←MD	8	
Turn off the device	0x50	GW→MD	8	
	0x50	GW←MD	8	
Clear/Delete all memory	0x52	GW→MD	8	
	0x52	GW←MD	8	
Notification for entering communication mode	0x54	GW←MD	8	
Read data	0x71	GW→MD	7	
	0x71	GW←MD	34	
Set user profile in the device	0x72	GW→MD	12	
	0x72	GW←MD	12	

2.2.1. [0x23] Read device clock time (4 bytes)

Message Name:	**Read device clock time (4 bytes)**
Message ID:	**0x23**
Message Description:	**Read device date and time (to minute)**
Src / Des:	**GW / MD**
Length:	**Request: 8 bytes / Response: 8 bytes**

Please refer to "device clock time" section for the format of date and time.

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x23	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x23	Day + Month + Year		Minute	Hour

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x23	GW commands MD to response device clock time
	DATA	4	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x23	MD response device clock time
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word)
	DATA	2		Data_2 : Minute, Data_3 : Hour

2.2.2. [0x24] Read device model

Message Name:	**Read device model**
Message ID:	**0x24**
Message Description:	**Read device model / project code**
Src / Des:	**GW / MD**
Length:	**Request: 8 bytes / Response: 8 bytes**

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read System Clock time	GW→MD	0x24	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x24	Project Code			

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x24	GW commands MD to response device model
	DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x24	MD response device clock time
	DATA	2		Data_1 + Data_0 : Device model in four digits (Data_1 is the higher byte)
	DATA	2		Data_3, Data_2 : undefined

2.2.3. [0x27] Read device serial number, part 1

Message Name:	Read the device serial number, part 1
Message ID:	0x27
Message Description:	Read the device serial number, part 1. This command cooperates with 0x28 to complete the serial number.
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) :
SN_7 + SN_6 + … + SN_0 (example: 32 50 21 02 40 13 10 2A)

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3

Read device serial number, part 1	GW→MD	0x27	0x0	0x0	0x0	0x0
Response from device	GW←MD	0x27	SN_0	SN_1	SN_2	SN_3

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x27	GW commands MD to response the device serial number, part 1
	DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x27	Read the device serial number, part 1
	DATA	4		Data_0 ~ Data_3 : SN_0 ~ SN_3

2.2.4. [0x28] Read device serial number, part 2

Message Name:	Read the device serial number, part 1
Message ID:	0x28
Message Description:	Read the device serial number, part 2. This command cooperates with 0x27 to complete the serial number.
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

The Serial Number for the device is composed of 16 characters (0~9, A~F) :
SN_7 + SN_6 + … + SN_0 (example: 32 50 21 02 40 13 10 2A)

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read device serial number, part 2	GW→MD	0x28	0x0	0x0	0x0	0x0
Response from device	GW←MD	0x28	SN_4	SN_5	SN_6	SN_7

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x28	GW commands MD to response the device serial number, part 2
	DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x28	Read the device serial number, part 2
	DATA	4		Data_0 ~ Data_3 : SN_4 ~ SN_7

2.2.5. [0x2B] Read storage number of data

Message Name:	Read storage number of data
Message ID:	0x2B
Message Description:	Read data counts
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Read Storage Number of Stored Readings	GW→MD	0x2B	0x0	0x0	0x0	0x0
Response from device	GW←MD	0x2B	Storage Number		0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x2B	GW commands MD to response the Number of Stored Readings
	DATA	4	0x0	

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x2B	MD response the the Number of Stored Readings
	DATA	2		Data_1+Data_0 = Storage Number (Word)

2.2.6. [0x33] Write System Clock time (4 bytes)

Message Name:	Write device clock time (4 bytes)
Message ID:	0x33
Message Description:	Write the device date and time (to minute)
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Please refer to "device clock time" section for the format of date and time.

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Write System Clock time	GW→MD	0x33	Day+Month+Year		Minute	Hour
Response from device.	GW←MD	0x33	Day+Month+Year		Minute	Hour

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x33	GW commands MD to write System Clock time (4 bytes)
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word)
	DATA	2		Data_2 : Minute, Data_3 : Hour

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x33	MD write System Clock time (4 bytes) O.K.
	DATA	2		Data_1 + Data_0 : Day+Month+Year (word)

	DATA	2		Data_2 : Minute, Data_3 : Hour

2.2.7. [0x50] Turn off the device

Message Name:	Turn off the device
Message ID:	0x50
Message Description:	Turn off the device from GW
Src / Des:	GW / MD
Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Turn off the device	GW→MD	0x50	0x0	0x0	0x0	0x0
	GW←MD	0x50	0x0	0x0	0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x50	GW commands MD to turn off the power.
	DATA	4	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x55	MD responses.
	DATA	4	0x0	NA

2.2.8. [0x52] Clear/Delete all Memory

Message Name:	Clear/Delete all memory
Message ID:	0x52
Message Description:	Delete all measurement data in the device
Src / Des:	GW / MD

Length:	Request: 8 bytes / Response: 8 bytes

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Clear/Delete all Memory Storages	GW→MD	0x52	0x0	0x0	0x0	0x0
Response from device.	GW←MD	0x52	0x0	0x0	0x0	0x0

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x52	GW commands MD to Clear/Delete all Memory Storages in Device EEPROM
	DATA	4	0x0	NA

#	Field	Size (Bytes)	Value	Description.
	ACK	1	0x52	Clear/Delete all Memory Storages in Device EEPROM O.K.
	DATA	4	0x0	NA

2.2.9. [0x54] Notification for entering communication mode

Message Name:	Notification for entering communication mode
Message ID:	0x54
Message Description:	Some models will send 0x54 command to host as a notification to indicate meter is ready for command from host. Host has to deal with this kind of notification in the beginning of connection (like RS-232) and prevent it to be mixed with the expect command response in receiving queue of host.
Src / Des:	MD / GW
Length:	Request: 8 bytes / Response: NA

Format

Message name	Direction	CMD	Data_0	Data_1	Data_2	Data_3
Notification of entering communication mode	GW←MD	0x54	0x0	0x0	0x0	0x0
		No need to response from GW side				

Field Description

#	Field	Size (Bytes)	Value	Description.
	CMD	1	0x54	MD reports to GW that it enters communication mode.
	DATA	4	0x0	NA

2.2.10. [0x71] Read Data

Message Name:	Read data
Message ID:	0x71
Message Description:	Read measurement data
Src / Des:	GW / MD
Length:	Request: 7 bytes / Response: 34 bytes

Format

	Byte	2	3	4	5	6	7
Message name	Direction	CMD	Length	Idx_L	Idx_H	Stop	CheckSum
Read data	GW→MD	0x71	0x02	Low_index	High_index	A3	[1..6]

	Byte	2	3	4~32	33	34
Message name	Direction	CMD	Length	Data	Stop	ChekSum
Response from device	GW←MD	0x71	0x1D	Hex	A5	[1…33]

Field Description

#	Field	Size (Bytes)	Value	Description

	CMD	1	0x71	Retrieve data in device with data index
	DATA	2	HEX	Idx_H + Idx_L = index of the data (word), start from 0. The value zero means newest record.

#	Field	Size (Bytes)	Value	Description
	ACK	1	0x71	Response data from MD to GW
	Length	1	0x1D	Length of the data bytes
4	Stable time	1		This field is only for Test N'GO Scale 550. Stable time. This unit is second and value range is form 0 to 255.
5	Year	1		Year, the year in A.D. format
6	Month	1		Month
7	Day	1		Day
8	Hour	1		Hour, in 24-hour format
9	Minute	1		Minute
10	Code	1		Code / User ID, start from 1
11	Gender	1		Gender. Female = 0, Male =1
12	Height(cm)	1		Height in cm
13	Height(in)	2		Height in inch x10. Byte 13 is the higher byte; Byte 14 is the lower byte
15	Age	1		Age, integer value
16	Cal unit	1		Calibration unit which means the current unit of scales. Kg=0, lb=1, st=2
17	Weight(kg)	2		Weight in kg x10. Byte 17 is the higher byte
19	Weight(lb)	2		Weight in lb x10. Byte 19 is the higher byte
21	BMI	2		BMI % x10. Byte 21 is the higher byte
23	BMR	2		Basal Metabolic Rate, in Kcal per day. Byte 23 is the higher byte
25	Bf	2		Body Fat. Bf% x10. Byte 25 is the higher byte
27	Bm	2		Body muscle. Bm% x10. Byte 27 is the higher byte
29	Bn	1		Body bone. Bn% x10.
30	Bw	2		Body water. Bw% x10. Byte 30 is the higher byte
32	Status	1		Only valid for body fat model. Severely Underweight=1, Underweight=2, Normal=3,

				Overweight=4, Obese=5

2.2.11. [0x72] Set user profile in the device

Message Name:	Set user profile in the device
Message ID:	0x72
Message Description:	Set user profile from GW to MD
Src / Des:	GW / MD
Length:	Request: 12 bytes / Response: 12 bytes

Format

	Byte	2	3	4~10	11	12
Message name	Direction	CMD	Length	Data	Stop	ChekSum
Set user profile	GW→MD	0x72	0x07	Hex	A3	[1…11]

	Byte	2	3	4~10	11	12
Message name	Direction	CMD	Length	Data	Stop	ChekSum
	GW←MD	0x72	0x07	Hex	A5	[1…11]

Field Description

#	Field	Size (Bytes)	Value	Description
2	CMD	1	0x72	Set user profile in the device.
3	Length	1	0x07	Length of the data segment
4	Code	1		Code / User ID, start from 1.
5	Gender	1		Gender. Female = 0, Male =1
6	Height(cm)	1		Height in cm. You have to input both in cm and in inch at the same time.
7	Height(in)	2		Height in inch. Byte 7 is the higher byte; Byte 8 is the lower byte
9	Age	1		Age, integer value
10	--- ---	1		No function

#	Field	Size (Bytes)	Value	Description
2	ACK	1	0x72	Response data from MD to GW
3	Length	1	0x07	Length of the data bytes
4	Code	1		Code / User ID, start from 1.
5	Gender	1		Gender. Female = 0, Male =1
6	Height(cm)	1		Height in cm
7	Height(in)	2		Height in inch. Byte 13 is the higher byte; Byte 14 is the lower byte
9	Age	1		Age, integer value
10	--- ---	1		--- ---

Data Format

2.2.12. Device clock time (4 bytes)

Direction	CMD/ ACK	Data_0	Data_1	Data_2	Data_3	Description
GW→MD	0x23	0x0	0x0	0x0	0x0	Read device date and time
GW←MD	0x23	Day+Month+Year (ref:TableA)		Minute+Hour (ref:TableB)		Response from device

Table A: Day + Month + Year

MSB **LSB**

<table>
<tr><td colspan="8">Byte 1</td><td colspan="8">Byte 0</td></tr>
<tr><td colspan="7">Year (7-bit)</td><td colspan="4">Month (4-bit)</td><td colspan="5">Day (5-bit)</td></tr>
<tr><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td><td>3</td><td>2</td><td>1</td><td>0</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
</table>

Table B: Minute and Hour

MSB **LSB**

<table>
<tr><td colspan="8">Byte 3</td><td colspan="8">Byte 2</td></tr>
<tr><td colspan="3"></td><td colspan="5">Hour (5-bit)</td><td></td><td></td><td colspan="6">Minute (6-bit)</td></tr>
<tr><td></td><td></td><td></td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td><td></td><td></td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr>
</table>